

5 INVENTIONS THAT WILL SAVE THE EARTH



From harnessing the power that fuels the sun, to little black boxes of wonder; we look at 5 inventions from the cutting edge of green technology that could one day save our planet

Siddharth Parwatay
siddharth.parwatay@thinkdigit.com

Going by some of their responses, the lovelies at beauty pageants seem to be bang on target about crucial issues facing the planet. Most flutter their eyebrows prettily about world peace (if that won't stop militancy, nothing will); protecting the environment (will give up furs and ammonia-based hair colour); and definitely doing something about global warming. Now this last one is where the leggy ladies from exotic lands have caught our attention. In reality, besides not participating in the swimsuit rounds – a major contributor to global warming – there's little that these lovelies can do. But let us take the beauties seriously for once, and find out from the geeks if their wishes can really come true.

Mankind's unquenchable thirst for energy is one of the biggest contributors to our planet's demise. We need energy to drive our cars, to power our computers, to light our streets and ultimately, power our lives. Therefore finding a super-efficient, non-polluting, sustainable method of producing this elixir is the answer. When it

comes to saving the earth, other than birth control, ironically it will be technology that will be the saviour. Could mankind's seemingly endless search for the perfect energy source be just around the corner? Maybe a clean, green and environmentally friendly technology that might even reverse the effects of pollution on our planet. Or a new source of water – despite being in abundance on the planet, water that's consumable is still scarce inland. Let's don our Nostradamus hats, peer into the future and find five inventions that could one day save our planet.

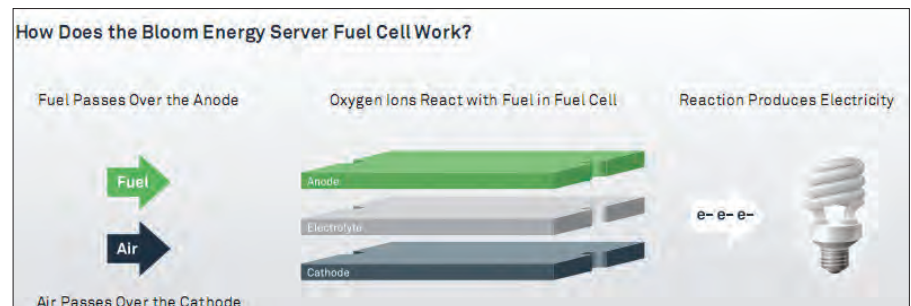
Bloombox

After almost a decade in complete secrecy, on a sunny day in silicon valley

recently, a nondescript scientist of Indian origin had the world's spotlight turned on him. In an almost unbelievable interview for prime time television show *60 Minutes*, Dr. K R Sridhar held up two little boxes about the size of large Rubik's cubes and claimed they could power an American home 24 hours a day. (He even joked that the same would power six homes back in India). This set into motion one of the most trending topics on the internet all of February and early March. From environmentalists, skeptics, to technologists; everyone was debating whether to believe Dr. K R Sridhar – the man whose invention promises to reinvent the energy business. Sridhar who worked for NASA prior to setting up Bloom Energy, has come up with a new innovation based on fuel cells. For those who are completely unaware of fuel cell technology, let's look at how they work. Unlike energy produced by combustion, fuel cells produce energy by a chemical reaction. A fuel (usually hydrogen) and an oxidant is pumped into a fuel cell which contains an electrolyte. A chemical reaction occurs that results in the creation of electricity, water and some amount of carbon dioxide as byproducts. Fuel cells are of different types and depending on the type of cell a significant amount of heat is also generated. This heat can also be utilised to increase the efficiency of energy production.

In common parlance Fuel Cells are most often associated with green cars. But their application in the mobile arena faced many challenges including carrying enough hydrogen for long journeys. The technology was dismissed by many as technically challenged, but Bloom has once again brought into the forefront.

The BloomBox is a fixed (stationary)



How the bloombox works